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Geometry–Frequency Dual-Domain Mamba–Transformer Network for Hyperspectral Unmixing

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***Abstract*—Hyperspectral unmixing jointly estimates spatial abundances and endmember spectra from mixed observations. However, mixed materials often exhibit heterogeneous spatial layouts and scale-varying structures, making it difficult to maintain region-level abundance coherence while preserving fine material boundaries. Meanwhile, reliable endmember recovery requires representing material-specific spectral variations together with local continuity across adjacent bands. To address these challenges, this paper proposes the Geometry–Frequency Dual-Domain Mamba–Transformer Network (GFMT-Net). In the geometry domain, Geometry-Adaptive Convolution (GAC) learns location-dependent vertical and horizontal sampling scales, allowing feature extraction to accommodate heterogeneous spatial structures. Global–Local Mamba–Transformer (GLMT) then applies Mamba to the complete spatial token sequence before refining local relations with windowed self-attention, promoting region-level abundance coherence while retaining fine spatial details. The reconstruction produced in the geometry domain is subsequently passed to the frequency domain, where the Fourier Spectral Encoder (FSE) enriches fixed-dimensional per-band descriptors with sinusoidal and nonperiodic basis responses and couples adjacent wavelengths through a one-dimensional convolution, thereby strengthening spectral representation while preserving local wavelength continuity. Experimental results on three real datasets demonstrate that GFMT-Net achieves competitive unmixing performance compared with state-of-the-art methods, particularly in abundance estimation.**

***Index Terms*—Hyperspectral unmixing, geometry-adaptive convolution, Fourier spectral encoding, state space models, Mamba–Transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) record a scene in hundreds of contiguous narrow bands, providing detailed reflectance signatures for material identification [1], [2]. This capability supports applications such as precision agriculture, mineral mapping, environmental monitoring, and urban analysis. However, the spatial resolution of a hyperspectral sensor is limited by the energy available within each narrow band, and a single pixel may therefore contain several materials. Hyperspectral unmixing addresses this mixed-pixel problem by decomposing each observed spectrum into a set of pure material signatures, termed endmembers, and their corresponding fractional abundances [3], [4]. Reliable unmixing consequently requires both spectrally faithful endmember recovery and spatially coherent abundance estimation.

Traditional unmixing methods formulate this inverse problem through geometric, statistical, or sparsity-based principles [3], [5]. Pure-pixel algorithms such as Pixel Purity Index [6], Vertex Component Analysis [7], and N-FINDR [8] identify endmembers from the extreme points of a data simplex. Bayesian models incorporate prior distributions to jointly infer endmembers and abundances [9], [10], whereas sparse regression and constrained matrix factorization exploit spectral libraries or abundance regularity [11], [12], [13], [14]. These approaches retain useful physical interpretations, but their performance depends on assumptions concerning pure pixels, mixing statistics, spectral libraries, or predefined regularizers. Such assumptions can be difficult to satisfy when materials are highly mixed or their spatial distributions are heterogeneous.

Deep learning provides a data-driven alternative by embedding the mixing model and physical abundance constraints into trainable architectures [15]. Autoencoder-based methods established a common formulation in which an encoder estimates latent abundances and a decoder reconstructs the observed spectra. Palsson et al. [16] proposed a neural network autoencoder constrained by the linear mixing model and showed that using spectral angle distance as the reconstruction objective improves robustness to noise. Ozkan et al. [17] subsequently introduced EndNet, which replaces the conventional inner-product projection with a spectral-angle-based projection and combines spectral similarity with sparsity penalties, producing more discriminative projections and sparser latent abundance estimates. Su et al. [18] developed the Deep Autoencoder Network, in which stacked autoencoders provide an informative initialization and a variational autoencoder jointly estimates endmembers and abundances under physical abundance constraints, thereby improving robustness to outliers and low signal-to-noise conditions. Although these methods progressively strengthen data-driven endmember and abundance estimation, their pixelwise or fully connected encoders make limited use of spatial correlations, and interpreting learned

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decoder parameters as endmembers can make the estimates sensitive to the optimization trajectory and initialization.

Convolutional neural networks (CNNs) extend this framework by exploiting local spatial context. Palsson et al. [19] introduced a convolutional autoencoder that aggregates neighboring pixels to learn spectral–spatial abundance representations. Zhao et al. [20] subsequently developed a 3-D-CNN-based unmixing framework whose encoder jointly learns spectral–spatial priors and whose decoder accounts for endmember variability, allowing neighborhood information and material-dependent spectral variation to be considered together. Moving toward unmixing-specific spatial designs, Tao et al. [21] proposed the Abundance-Guided Spectral and Spatial Network (A2SN), which employs different kernels to capture spectral and spatial information and extracts endmember features directly from the network, reducing its reliance on externally initialized decoder weights. Xu et al. [22] further proposed the Multiscale Convolutional Mask Network (MsCM-Net), which combines multiscale convolution with mixed-region masking to exploit spatial structures at different ranges while suppressing interference from highly mixed regions. These developments demonstrate the value of spatial context for unmixing. Nevertheless, most convolutional operators sample every location with the same square grid. Their fixed geometry can be poorly matched to anisotropic material distributions and irregular boundaries, while stacking local convolutions alone provides only an indirect route to scene-wide interactions.

Self-attention provides a content-adaptive means of modeling spatial relations beyond fixed convolutional neighborhoods [23]. Ghosh et al. [24] proposed a convolutional autoencoder–Transformer network in which multihead self-patch attention captures nonlocal dependencies among encoded image patches, improving the estimation of both abundances and endmembers. Moving toward spatially localized attention, Yang et al. [25] developed UST-Net, which computes self-attention within local windows and shifts the window partitions to exchange information across neighboring regions. Combined with U-shaped multiscale sampling, this design produces context-rich abundance representations while retaining local details. Su et al. [26] further showed that local continuity is not automatically preserved by patch-based vision Transformers. Their Self-Supervised Relation-Aware Vision Transformer (SRViT) combines a self-embedded encoder with pixel-level positional encoding to mitigate the disruption of fine-grained spatial relations caused by nonoverlapping tokenization. These studies show that the effective spatial range of self-attention depends strongly on its tokenization and attention organization: global patch interactions support nonlocal modeling, whereas windowed or locally aware designs emphasize relations within compact regions. However, global self-attention scales quadratically with the number of spatial tokens, while localizing attention reduces cost but limits direct interaction across regions. Achieving broad spatial context while controlling computation and preserving pixel-level continuity therefore remains challenging.

Selective state space models (SSMs) provide an efficient route to long-range dependency modeling. Gu and Dao [27] introduced Mamba, whose input-dependent selection enables content-aware sequence propagation with linear complexity, and Liu et al. [28] extended this principle to two-dimensional visual features through multidirectional scanning. Moving from general vision to HSI classification, Li et al. [29] proposed MambaHSI, which combines pixel-level spatial Mamba modeling with grouped spectral modeling and adaptive fusion. Pan et al. [30] developed MambaLG to integrate local and global spatial context with short- and long-range spectral relations, whereas Liang et al. [31] introduced DBMLLA, coupling spatial and spectral Mamba-like linear attention with depthwise separable convolution for local refinement. Recent studies have extended Mamba to blind hyperspectral unmixing. Chen et al. [32] proposed UNMamba, which cascades spatial and spectral Mamba blocks for abundance estimation and learns endmembers from trainable random sequences without VCA initialization. Zhang et al. [33] developed MBUNet, using CNNs and bidirectional Mamba within dual spatial–spectral streams to coordinate local features and image-level dependencies. Zhao et al. [34] introduced FACM, which performs frequency-adaptive abundance estimation and endmember extraction and constrains reconstruction through Fourier magnitude and phase. These studies demonstrate that Mamba can support efficient spatial–spectral modeling in both classification and unmixing [35]. However, existing designs largely organize spatial interactions through predefined scans, cascades, or parallel branches. Adapting the spatial support to irregular material layouts while combining scene-wide propagation with explicit local refinement therefore remains an open problem.

Alongside advances in abundance estimation, endmember recovery has increasingly benefited from richer models of spectral information. Borsoi et al. [36] proposed the Deep Generative Unmixing algorithm (DeepGUn), which trains variational autoencoders on pure pixels to learn low-dimensional manifolds of endmember variability and embeds the resulting generative models into the linear mixing model. This approach provides flexible representations of material signatures while retaining a physically interpretable mixing process. From a transform-domain perspective, Xu et al. [37] proposed SWC-Net, which decomposes an HSI into low- and high-frequency components through the stationary wavelet transform and processes them with parallel convolutional autoencoder branches, allowing complementary structural and detailed information to contribute to unmixing. However, these approaches focus primarily on either latent spectral variability or multiscale transform features. How to convert the spatial evidence carried by individual bands into expressive endmember representations while explicitly exploiting local dependence along the ordered wavelength sequence remains less explored.

To address these issues, we propose the Geometry–Frequency Dual-Domain Mamba–Transformer Network (GFMT-Net). In the geometry domain, Geometry-Adaptive Convolution (GAC) learns location-dependent vertical and horizontal sampling scales to accommodate heterogeneous spatial structures, and a Global–Local Mamba–Transformer (GLMT) applies Mamba to the complete spatial token sequence before refining local relations with windowed self-

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attention to estimate abundances. The resulting abundances produce an internal reconstruction that serves as the explicit interface to the frequency-domain endmember branch. The Fourier Spectral Encoder (FSE) then adaptively pools every reconstructed band to a fixed spatial grid, enriches the resulting per-band descriptors with sinusoidal and nonperiodic basis responses, and uses a one-dimensional convolution to couple adjacent wavelengths. This sequential design provides an expressive endmember representation whose parameter dimensions are independent of the input spatial dimensions. The main contributions are summarized as follows.

1) We develop GFMT-Net, a coupled geometry–frequency dual-domain framework for hyperspectral unmixing. It links geometry-domain abundance estimation to frequency-domain endmember recovery through an internal reconstruction, enabling spatial structure modeling and spectral representation to be coordinated within a unified estimation process.
2) In the geometry domain, we introduce GAC as the first geometry-adaptive convolution designed for hyperspectral unmixing. GAC learns location-dependent vertical and horizontal scales for an axis-aligned sampling grid, allowing abundance feature extraction to accommodate heterogeneous material layouts and irregular boundaries.
3) Building on the geometry-adaptive features produced by GAC, we design GLMT as the first Mamba–Transformer backbone for hyperspectral unmixing. GLMT applies Mamba to the complete spatial token sequence before performing self-attention within local windows. This ordering propagates information across window boundaries before refining fine-grained spatial relations, supporting scene-wide abundance consistency under controlled attention cost.
4) In the frequency domain, we develop FSE for reconstruction-conditioned endmember estimation. FSE adaptively pools each reconstructed band to a fixed spatial grid, enriches the resulting descriptors with sinusoidal and nonperiodic basis responses, and couples adjacent wavelengths through a one-dimensional convolution, enabling expressive endmember representation with parameter dimensions independent of the input spatial dimensions.

The remainder of this paper is organized as follows. Section II reviews related works. Section III details the architecture and components of GFMT-Net. Section IV presents the experimental evaluation and ablation studies. Section V concludes the paper and discusses future research directions.

II. Related Work

This section reviews the two algorithmic foundations of GLMT: selective state space modeling for efficient sequence processing and window-based self-attention for explicit local pairwise interactions. The task-specific organization of these operators is described in Section III-C.

A. State Space Models

SSMs constitute a classical family of sequence models that propagate information through recursive hidden-state transitions [38]. To process a discrete input sequence $\{\mathbf{x}_t\}_{t=1}^{T}$, where T denotes a generic sequence length, an SSM is implemented in discrete form as

$$\mathbf{h}_t = \bar{\mathbf{A}}\mathbf{h}_{t-1} + \bar{\mathbf{B}}\mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C}\mathbf{h}_t, \tag{1}$$

where $\mathbf{h}_t$ is the hidden state and $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$, and $\mathbf{C}$ are the discrete state-transition, input-projection, and output-projection matrices, respectively. When the model is derived from a continuous-time SSM, $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ can be obtained using a discretization scheme such as zero-order hold. Structured SSMs exploit the equivalence between the recurrence and a structured convolution with a kernel of length T for efficient parallel evaluation.

In a conventional structured SSM, the transition and projection parameters are shared across sequence positions. Mamba [27] introduces a selective mechanism that makes the input projection, output projection, and discretization step dependent on the current token:

$$\begin{aligned} \mathbf{B}_t &= \mathrm{Linear}_{\mathbf{B}}(\mathbf{x}_t), \\ \mathbf{C}_t &= \mathrm{Linear}_{\mathbf{C}}(\mathbf{x}_t), \\ \Delta_t &= \mathrm{Softplus}(\mathrm{Parameter} + \mathrm{Linear}_{\Delta}(\mathbf{x}_t)). \end{aligned} \tag{2}$$

The resulting selective scan conditions the state update on the input while retaining linear computational complexity with respect to T for fixed state size. Unlike self-attention, information is propagated through state transitions rather than an explicitly materialized pairwise similarity matrix.

For visual inputs, the generic sequence index t must be associated with positions in a two-dimensional feature map. A feature map $\mathbf{X} \in \mathbb{R}^{H\times W\times d}$ is commonly serialized into HW spatial tokens, each containing a d-dimensional feature vector; the processed sequence is then restored to its two-dimensional layout. Because the state recurrence follows this serialization, the traversal order determines which locations are consecutive and how many state transitions separate them. Consequently, neighbors in the two-dimensional map may become distant in the one-dimensional sequence. Visual SSMs have therefore explored multidirectional traversals to reduce directional bias and improve spatial coverage [28]. The spatial extent and ordering of the sequence thus directly affect information propagation.

B. Self-Attention Mechanism

Self-attention is the computational primitive of the Transformer architecture [39]. Given N input tokens $\mathbf{X} \in \mathbb{R}^{N\times d}$, learned projections produce queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$. Scaled dot-product attention then computes

$$\mathrm{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_k}}\right)\mathbf{V}, \tag{3}$$

where d_k is the query and key dimension. The resulting $N \times N$ attention matrix explicitly represents pairwise compatibility between tokens.

Multi-head self-attention (MHSA) applies this operation in h learned subspaces and concatenates the resulting heads:

$$\begin{aligned} \mathrm{MHSA}(\mathbf{X}) &= \mathrm{Concat}(\mathrm{head}_1,\ldots,\mathrm{head}_h)\mathbf{W}_O, \\ \mathrm{head}_i &= \mathrm{Attention}(\mathbf{X}\mathbf{W}_Q^i, \mathbf{X}\mathbf{W}_K^i, \mathbf{X}\mathbf{W}_V^i), \end{aligned} \tag{4}$$

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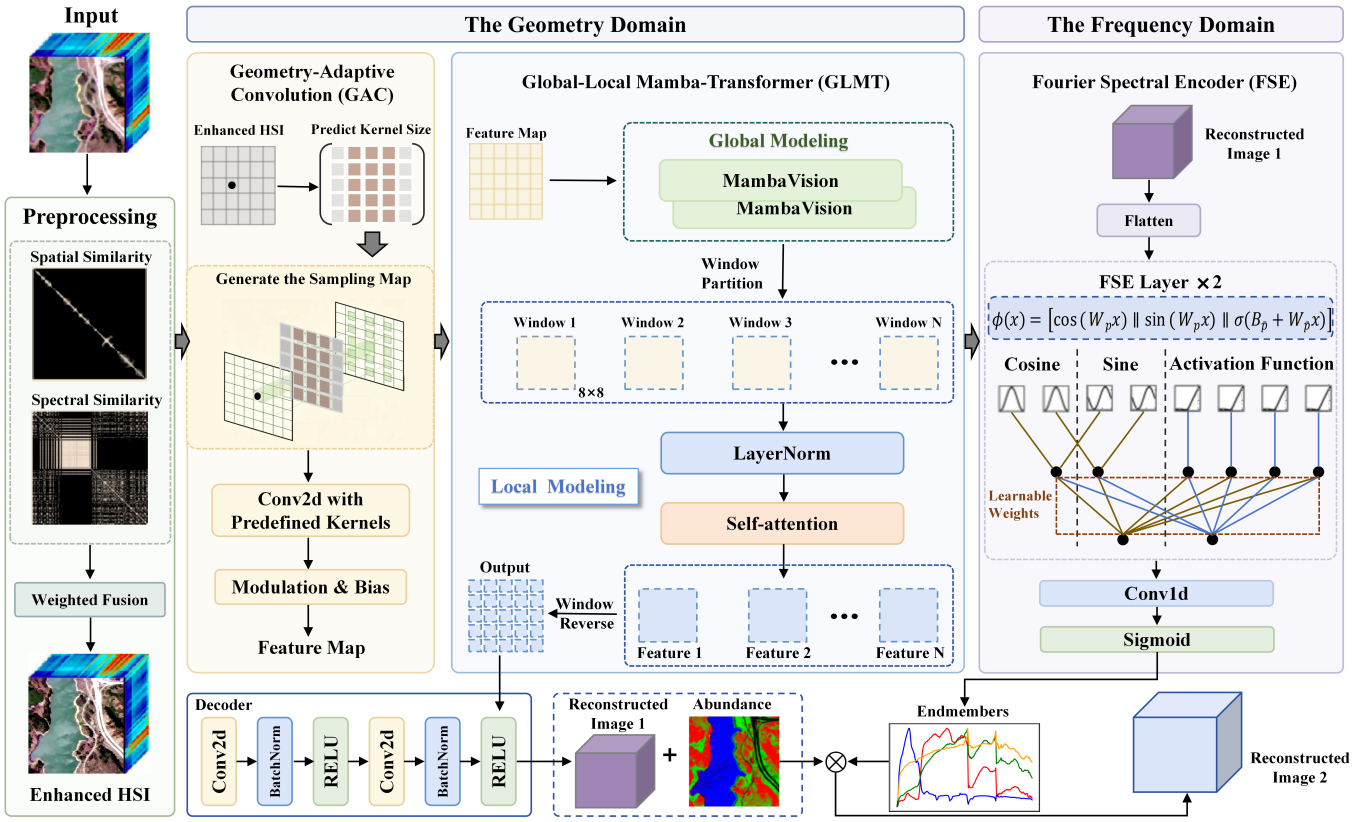

Fig. 1. Overall architecture of the proposed GFMT-Net. Shared similarity-enhanced preprocessing feeds the geometry pathway, where GAC and GLMT estimate abundances and produce an internal reconstruction. Conditioned on this reconstruction, the frequency pathway uses FSE layers and a wavelength-axis convolution to estimate endmembers.

where $\mathbf{W}_Q^i$, $\mathbf{W}_K^i$, $\mathbf{W}_V^i$, and $\mathbf{W}_O$ are learned projections. Standard MHSA has $\mathcal{O}(N^2 d)$ computational complexity and $\mathcal{O}(N^2)$ memory cost because it materializes pairwise scores over the complete sequence.

Window-based self-attention reduces this cost by applying MHSA independently to nonoverlapping $W_s \times W_s$ regions. Its per-window complexity is $\mathcal{O}(W_s^4 d)$, and its total complexity over an $H \times W$ feature map is $\mathcal{O}(HWW_s^2 d)$ for a fixed window size. Swin Transformer [40] alternates regular and cyclically shifted window partitions so that information can propagate between windows across successive layers. In contrast, the windowed stage of GLMT uses fixed nonoverlapping windows without shifted partitioning; cross-window information is supplied by the complete-sequence MambaVision stage that precedes window partitioning. Selective state-space processing and window-based self-attention therefore operate over different spatial extents, while their task-specific organization is detailed in Section III-C.

III. Proposed Method

In this section, we detail the architecture of GFMT-Net. Throughout the section, B, L, H, W, and P denote the batch size, number of spectral bands, input height, input width, and number of endmembers, respectively; C, H', and W' denote the channel and spatial dimensions of the latent feature map. We also use $N_p = HW$ for the number of pixels and $N_s = k_h k_w$ for the number of GAC sampling points. Bold uppercase symbols denote matrices or tensors, whereas lowercase symbols denote scalars or vectors unless otherwise specified.

As illustrated in Fig. 1, the input HSI $\mathbf{Y}$ first undergoes similarity-enhanced preprocessing, which propagates spectral and spatial affinities to produce the shared representation $\mathbf{Y}'$. In the geometry pathway, patch embedding maps $\mathbf{Y}'$ to a latent feature map, GAC adapts the spatial sampling support through location-dependent vertical and horizontal scales, and GLMT successively performs complete-sequence MambaVision modeling and windowed self-attention. The resulting geometry features are upsampled and delivered to separate abundance and reconstruction heads, yielding the abundance tensor $\mathbf{A}$ and the internal reconstruction $\mathbf{Y}_r$, respectively. The internal reconstruction explicitly connects the two domains: in the frequency pathway, adaptive pooling converts each reconstructed band into a fixed-dimensional spatial descriptor, two FSE layers generate periodic and nonperiodic basis responses, and a one-dimensional convolution along the wavelength axis produces the endmember tensor $\mathbf{E}$. Here, "Fourier" denotes learnable cosine and sine basis mappings on each pooled band descriptor rather than an FFT or DFT along the wavelength axis; adjacent-wavelength coupling is instead provided by the subsequent one-dimensional convolution. Finally, $\mathbf{A}$ and $\mathbf{E}$ are combined through the linear mixing model to form the final reconstruction $\hat{\mathbf{Y}}_f$.

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For clarity, the principal tensor interfaces are summarized as follows. A batched input $\mathbf{Y} \in \mathbb{R}^{B\times L\times H\times W}$ is transformed by similarity-enhanced preprocessing into $\mathbf{Y}'$ with the same dimensions. Patch embedding maps $\mathbf{Y}'$ to $\mathbf{X}_0 \in \mathbb{R}^{B\times C\times H'\times W'}$, and GAC produces $\mathbf{X}_g$ with the same shape. GLMT processes $\mathbf{X}_g$ and preserves this latent shape; after bilinear upsampling, its output becomes $\mathbf{F} \in \mathbb{R}^{B\times C\times H\times W}$. The abundance head produces $\mathbf{A} \in \mathbb{R}^{B\times P\times H\times W}$, while the reconstruction head produces the internal reconstruction $\mathbf{Y}_r \in \mathbb{R}^{B\times L\times H\times W}$. FSE receives $\mathbf{Y}_r$, pools each of its L bands to a fixed $G\times G$ grid, and maps the resulting descriptors to $\mathbf{E} \in \mathbb{R}^{B\times L\times P}$. Finally, flattening the spatial dimensions of $\mathbf{A}$ gives $\mathbf{A}_f \in \mathbb{R}^{B\times P\times N_p}$, and the batchwise product $\mathbf{E}_b\mathbf{A}_{f,b}$ yields $\hat{\mathbf{Y}}_f \in \mathbb{R}^{B\times L\times N_p}$. Thus, $\mathbf{Y}_r$ is the interface from the geometry pathway to FSE, whereas $\mathbf{A}$ and $\mathbf{E}$ are the two factors used by the final linear mixing reconstruction.

A. Similarity-Enhanced Preprocessing

Hyperspectral images exhibit strong correlations along both the spectral and spatial dimensions, in that adjacent bands capture contiguous molecular absorption features, while neighboring pixels often belong to the same material. However, raw HSI inputs present these correlations in an unstructured form that can hinder subsequent feature extraction. To address this, we apply a similarity-based enhancement that explicitly encodes spectral and spatial affinities before the input enters the main network. This preprocessing step amplifies the intrinsic data structure, providing a more coherent foundation for GAC's adaptive receptive field learning.

For clarity, the preprocessing equations are written for one HSI and omit the batch dimension. Given $\mathbf{Y} \in \mathbb{R}^{L\times H\times W}$, we reshape it to $\mathbf{S} \in \mathbb{R}^{L\times N_p}$, where each column is a pixel spectrum. Two similarity matrices are then computed.

The spectral similarity matrix $\mathbf{M}_{\text{spec}} \in \mathbb{R}^{L\times L}$ captures correlations between spectral bands:

$$\mathbf{M}_{\text{spec}} = \exp\left(-\mathbf{D}_{\text{spec}}\right), \quad [\mathbf{D}_{\text{spec}}]_{i,j} = \|\mathbf{S}_{i,:} - \mathbf{S}_{j,:}\|_2^2, \tag{5}$$

where $\mathbf{S}_{i,:}$ denotes the i-th row of $\mathbf{S}$, representing the spatial response of band i. Thus, each entry of $\mathbf{D}_{\text{spec}}$ is calculated between two bands along the pixel axis, using all N_p spatial samples, and the resulting matrix has size $L\times L$. The exponential is applied elementwise.

The spatial similarity matrix $\mathbf{M}_{\text{spat}} \in \mathbb{R}^{N_p\times N_p}$ captures correlations between pixels:

$$\mathbf{M}_{\text{spat}} = \exp\left(-\mathbf{D}_{\text{spat}}\right), \quad [\mathbf{D}_{\text{spat}}]_{p,q} = \|\mathbf{S}_{:,p} - \mathbf{S}_{:,q}\|_2^2, \tag{6}$$

where $\mathbf{S}_{:,p}$ denotes the p-th column of $\mathbf{S}$, representing the spectrum of pixel p. Accordingly, each entry of $\mathbf{D}_{\text{spat}}$ is calculated between two pixels along the spectral axis, using all L bands, and the resulting matrix has size $N_p\times N_p$. The exponential is again applied elementwise.

The spectral and spatial propagation terms are first scaled independently to $[0,1]$ and then combined:

$$\begin{aligned}
\mathbf{S}_{\text{spec}} &= \text{norm}_{01}(\mathbf{M}_{\text{spec}}\mathbf{S}), \\
\mathbf{S}_{\text{spat}} &= \text{norm}_{01}(\mathbf{S}\mathbf{M}_{\text{spat}}), \\
\mathbf{S}' &= \text{norm}_{01}(\rho\mathbf{S}_{\text{spec}} + (1-\rho)\mathbf{S}_{\text{spat}}), \\
\mathbf{Y}' &= \text{reshape}_{L\times H\times W}(\mathbf{S}'),
\end{aligned} \tag{7}$$

where $\rho \in [0,1]$ controls the relative contribution of spectral and spatial propagation. Although $\mathbf{M}_{\text{spat}}$ is written as a dense matrix for notational clarity, the product $\mathbf{S}\mathbf{M}_{\text{spat}}$ is computed blockwise along the pixel dimension, and the complete affinity matrix is never materialized. This implementation is algebraically equivalent to the dense product while reducing peak GPU memory usage. For any intermediate matrix $\mathbf{Z} \in \mathbb{R}^{L\times N_p}$, $\text{norm}_{01}(\cdot)$ is defined as

$$\text{norm}_{01}(\mathbf{Z}) = \begin{cases} \dfrac{\mathbf{Z} - z_{\min}}{z_{\max} - z_{\min}}, & z_{\max} > z_{\min}, \\ \mathbf{0}, & z_{\max} = z_{\min}, \end{cases} \tag{8}$$

where $z_{\min} = \min_{\ell,n} Z_{\ell n}$ and $z_{\max} = \max_{\ell,n} Z_{\ell n}$. The minimum and maximum are therefore computed over all entries of each intermediate matrix independently for each HSI, rather than per band, per pixel, or across a batch. The zero-range branch prevents division by zero and returns a matrix of zeros with the same dimensions as $\mathbf{Z}$. The spectral term propagates information between bands according to their spatial-response similarity, whereas the spatial term propagates spectra between pixels according to their spectral similarity. Their weighted combination provides the common input to the geometry pathway.

B. Geometry-Adaptive Convolution

Standard CNNs commonly use fixed square sampling grids whose geometry does not vary across locations. We therefore use GAC after patch embedding. GAC predicts location-dependent continuous vertical and horizontal sampling scales for an axis-aligned rectangular grid. Its shared discrete grid dimensions are determined from spatially averaged predictions, while the continuous scales vary across latent spatial locations.

1) Height-Width Learner: The enhanced input $\mathbf{Y}' \in \mathbb{R}^{B\times L\times H\times W}$ is first mapped by patch embedding to $\mathbf{X}_0 \in \mathbb{R}^{B\times C\times H'\times W'}$. GAC predicts a continuous sampling height h_0 and width w_0 at each latent spatial location through two lightweight subnetworks f_{θ_1} and f_{θ_2}:

$$h_0 = f_{\theta_1}(\mathbf{X}_0), \quad w_0 = f_{\theta_2}(\mathbf{X}_0). \tag{9}$$

Each subnetwork consists of two convolutional layers with batch normalization and LeakyReLU activation, followed by a sigmoid output. The predicted $h_0, w_0 \in \mathbb{R}^{B\times 1\times H'\times W'}$ vary spatially, allowing different latent locations to use different sampling geometries.

2) Sampling Point Determination: The sigmoid responses are affinely scaled to the permitted continuous range before being used as h_0 and w_0. GAC then determines one pair of discrete grid dimensions for the current batch by averaging these continuous scales over the batch and latent spatial positions:

$$\bar{h} = \frac{1}{BH'W'}\sum_{b,u,v} h_0^{(b)}(u,v), \quad \bar{w} = \frac{1}{BH'W'}\sum_{b,u,v} w_0^{(b)}(u,v). \tag{10}$$

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The averages are converted to odd grid dimensions using a sampling-density factor s:

$$k_h = \psi\left(\left\lfloor \frac{\bar{h}}{s} \right\rfloor\right), \quad k_w = \psi\left(\left\lfloor \frac{\bar{w}}{s} \right\rfloor\right), \\ \psi(x) = \min(7, \max(3, x - [\![x \text{ is even}]\!])), \tag{11}$$

where $s = 1$ in our implementation and $\psi(\cdot)$ maps its argument to an odd integer in $[3, 7]$. The grid therefore contains $N_s = k_h k_w$ sampling points. During the first 100 epochs, (k_h, k_w) is updated from the batch-averaged predictions and is fixed thereafter; the continuous h_0 and w_0 remain location dependent and continue to determine the sampling offsets.

3) Sampling Map Generation: For each latent spatial position $\mathbf{p}_0(u, v) = (u, v)$, GAC constructs location-dependent sampling offsets. Let $\mathcal{G} \in \mathbb{R}^{k_h \times k_w \times 2}$ denote the centered base grid, whose (i, j)-th coordinate is

$$\mathbf{g}_{ij} = \left(g_{ij}^h, g_{ij}^w\right) = \left(\frac{2i - k_h - 1}{2}, \frac{2j - k_w - 1}{2}\right). \tag{12}$$

At batch index b and location (u, v), the corresponding offset and sampling coordinate are

$$\mathbf{r}_{buvij} = \left(\frac{h_0^{(b)}(u, v)}{k_h} g_{ij}^h, \frac{w_0^{(b)}(u, v)}{k_w} g_{ij}^w\right), \\ \mathbf{p}_{buvij} = \mathbf{p}_0(u, v) + \mathbf{r}_{buvij}. \tag{13}$$

The complete offset tensor is $\mathbf{R} \in \mathbb{R}^{B \times H' \times W' \times k_h \times k_w \times 2}$. Equation (13) scales the vertical and horizontal extents independently while retaining an axis-aligned grid. Because $\mathbf{p}_{buvij}$ is generally nonintegral, the feature at each sampling coordinate is obtained by bilinear interpolation.

After sampling, the $N_s = k_h k_w$ feature vectors at each latent position are rearranged into $k_h \times k_w$ blocks and tiled to form $\mathbf{S}_{\text{map}} \in \mathbb{R}^{B \times C \times (k_h H') \times (k_w W')}$.

4) Convolution and Modulation: A shared convolution with kernel size $k_h \times k_w$ and stride (k_h, k_w) is applied to the location-dependent sampled supports in $\mathbf{S}_{\text{map}}$. The convolutional response is then modulated by input-dependent multiplicative and additive maps:

$$\mathbf{X}_g = \text{Conv}_{\mathbf{W}}(\mathbf{S}_{\text{map}}) \odot \boldsymbol{\Gamma}(\mathbf{X}_0) + \boldsymbol{\beta}(\mathbf{X}_0), \tag{14}$$

where $\mathbf{W}$ contains the shared convolutional weights, and $\boldsymbol{\Gamma}(\mathbf{X}_0)$ and $\boldsymbol{\beta}(\mathbf{X}_0)$ are predicted modulation maps. The output $\mathbf{X}_g \in \mathbb{R}^{B \times C \times H' \times W'}$ has the same shape as $\mathbf{X}_0$ and is passed to GLMT.

C. Global–Local Mamba–Transformer

Following spatial adaptation by GAC, feature extraction is performed by GLMT. Its input is $\mathbf{X}_g \in \mathbb{R}^{B \times C \times H' \times W'}$, where $C = 64$ in our implementation. The C channels are learned from all input bands and are therefore spectrally informed latent features rather than the original spectral axis L. GLMT first applies two MambaVision mixers to the complete sequence of $H'W'$ spatial tokens and then restores the two-dimensional layout for refinement by two attention blocks within fixed nonoverlapping 8×8 windows. Both stages therefore model spatial interactions over different extents, while the spectral information remains encoded in each token's latent feature vector.

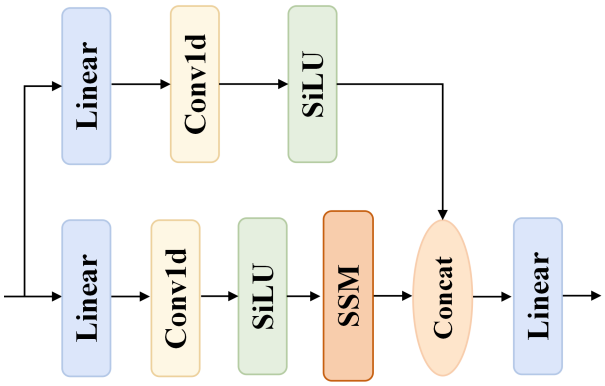

Fig. 2. Architecture of the MambaVision block. The block adapts the original design for hyperspectral feature processing by combining an SSM path with an SSM-free path.

Rather than applying the two operators directly to raw spectral vectors, GLMT processes the spectrally informed latent tokens produced by patch embedding and GAC. Processing the complete spatial sequence before window partitioning provides a route for information to reach tokens in different windows before local attention is applied.

1) Complete-Sequence MambaVision Stage: As shown in Fig. 1, the first stage consists of two MambaVision mixers applied to the complete latent feature map. Before each mixer, layer normalization is applied to the sequence representation. The output is then reshaped back to its two-dimensional spatial layout before the windowed blocks. This ordering provides cross-window spatial information to the tokens used by local attention; the extent to which this improves material discrimination is evaluated empirically rather than assumed from the architecture alone.

The features produced by the complete-sequence MambaVision stage are padded to multiples of the window size $W_s = 8$ and partitioned into nonoverlapping 8×8 windows. The two Transformer blocks process the resulting window-token sets, after which inverse partitioning restores the spatial layout. Attention cost is therefore bounded by the fixed number of tokens in each window, while the preceding MambaVision stage provides a route for information flow across window boundaries.

2) MambaVision: As shown in Fig. 2, the mixer follows the two-branch MambaVision design [41] and uses the selective SSM introduced by Mamba [27]. The latent input $\mathbf{X}_g \in \mathbb{R}^{B \times C \times H' \times W'}$ is reshaped to $\mathbf{X}_{\text{seq}} \in \mathbb{R}^{B \times (H'W') \times C}$. The sequence axis indexes spatial positions, whereas the feature axis contains latent channels derived jointly from the input spectrum.

After linear projection, the mixer splits the features into an SSM branch $\mathbf{x}$ and an SSM-free branch $\mathbf{z}$. Both branches apply depthwise one-dimensional convolution and SiLU activation:

$$\mathbf{x}' = \text{SiLU}(\text{DWConv1d}_x(\mathbf{x})), \\ \mathbf{z}' = \text{SiLU}(\text{DWConv1d}_z(\mathbf{z})). \tag{15}$$

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The $\mathbf{x}'$ branch is then processed by the selective state-space mechanism reviewed in Section II-A, producing $\mathbf{y}$, whereas $\mathbf{z}'$ bypasses the selective scan. The input-dependent transition and projection parameters condition each state update on the latent feature at the corresponding spatial token.

The two branches are concatenated rather than combined through multiplicative gating, and the result is projected back to dimension C:

$$\mathbf{Y}_{\text{mamba}} = \text{Proj}(\text{Concat}(\mathbf{y}, \mathbf{z}')) . \tag{16}$$

For fixed state dimension and expansion ratio, the complete-sequence selective scan scales linearly with the number of spatial tokens, $\mathcal{O}(H'W'C)$. The attention-score computation within one window scales as $\mathcal{O}(W_s^4 C)$, and the corresponding computation over all windows scales as $\mathcal{O}(H'W'W_s^2 C)$. These expressions use C, rather than L, because GLMT operates on latent channels after patch embedding.

3) Window Self-Attention: The Transformer blocks in the latter half of GLMT employ the MHSA formulation in Section II-B within each 8×8 window. For a windowed feature $\mathbf{X}_w \in \mathbb{R}^{W_s^2 \times C}$, four heads are used, giving a per-head dimension $d_k = C/4$; their outputs are concatenated and projected back to dimension C. Each block uses pre-normalized MHSA and a feed-forward network with residual connections. The windows remain fixed and nonoverlapping across both blocks, so the attention stage does not establish direct cross-window interactions through shifted partitions. Instead, the preceding complete-sequence MambaVision stage supplies the route for cross-window spatial information flow.

4) Abundance and Reconstruction Heads: After the windowed GLMT stage, the latent feature map is bilinearly upsampled to the input spatial resolution, forming $\mathbf{F} \in \mathbb{R}^{B \times C \times H \times W}$. Two lightweight convolutional heads then produce the abundance tensor and the internal reconstruction.

Abundance head: The feature map $\mathbf{F}$ is passed through two $k \times k$ convolutional layers with batch normalization and ReLU activation, progressively reducing the channel dimension from C to 32 and then to P. The final ReLU enforces abundance nonnegativity and produces $\mathbf{A} \in \mathbb{R}^{B \times P \times H \times W}$.

Reconstruction head: The reconstruction head maps $\mathbf{A}$ through two $k \times k$ convolutional layers that change the channel dimension from P to 48 and then to L. Both layers use batch normalization and ReLU in the current implementation, producing $\mathbf{Y}_r \in \mathbb{R}^{B \times L \times H \times W}$. The abundance tensor $\mathbf{A}$ is retained for the final linear-mixing reconstruction, while $\mathbf{Y}_r$ is passed to the endmember estimation module.

D. Fourier Spectral Encoder

FSE receives the internal reconstruction $\mathbf{Y}_r$ and estimates endmembers from its bandwise spatial evidence rather than interpreting decoder weights as endmember spectra. This choice changes the source from which endmembers are computed but does not eliminate the general dependence of neural layers on parameter initialization.

1) FSE Layer Design: A FSE layer decomposes its output into cosine, sine, and GELU-activated components. Given input $\mathbf{u} \in \mathbb{R}^{d_{\text{in}}}$ and output dimension d_{out}, a periodic ratio $p_r = 0.25$ controls the fraction of dimensions allocated to the trigonometric components:

$$d_p = \lfloor p_r \cdot d_{\text{out}} \rfloor, \quad d_g = d_{\text{out}} - 2 \cdot d_p, \tag{17}$$

where d_p dimensions are allocated to the cosine basis, d_p to the sine basis, and the remaining d_g dimensions to a nonperiodic component. The forward pass is

$$\begin{aligned} \mathbf{v} = \text{concat}\big(& \cos(\mathbf{W}_p \mathbf{u} + \mathbf{b}_p), \\ & \sin(\mathbf{W}_p \mathbf{u} + \mathbf{b}_p), \\ & \text{GELU}(\mathbf{W}_g \mathbf{u} + \mathbf{b}_g)\big), \end{aligned} \tag{18}$$

where $\mathbf{W}_p \in \mathbb{R}^{d_p \times d_{\text{in}}}$ and $\mathbf{W}_g \in \mathbb{R}^{d_g \times d_{\text{in}}}$ are learnable weight matrices, $\mathbf{b}_p$ is a learned phase offset, and $\mathbf{b}_g$ is the bias of the nonperiodic branch. The cosine and sine branches share the same affine projection, while the GELU branch supplies nonperiodic nonlinear features. This construction increases the variety of basis responses available to the encoder, but it does not by itself guarantee a better spectral fit.

2) Endmember Extraction Architecture: The parameter dimensions of the endmember branch are independent of the scene dimensions H and W. Given $\mathbf{Y}_r \in \mathbb{R}^{B \times L \times H \times W}$, adaptive average pooling first produces a fixed $G \times G$ output grid for every band:

$$\mathbf{U} = \text{vec}(\text{Pool}_{G \times G}(\mathbf{Y}_r)) \in \mathbb{R}^{B \times L \times G^2}. \tag{19}$$

Two band-shared FSE layers with $p_r = 0.25$ map the last dimension from G^2 to 100 and then to 40. These layers act independently on the pooled spatial descriptor of each band and do not perform a Fourier transform along the wavelength axis. The resulting tensor is transposed to $B \times 40 \times L$ and processed by a one-dimensional convolution of kernel size 5 along the ordered wavelength axis. A sigmoid bounds the output to $[0, 1]$, and transposition yields $\mathbf{E} \in \mathbb{R}^{B \times L \times P}$. We set $G = 10$. Adaptive pooling removes parameter dependence on HW, while the subsequent spectral convolution explicitly couples adjacent bands.

After reshaping the abundance tensor to $\mathbf{A}_f \in \mathbb{R}^{B \times P \times N_p}$, the final LMM reconstruction is computed batchwise as

$$\hat{\mathbf{Y}}_{f,b} = \mathbf{E}_b \mathbf{A}_{f,b}, \qquad b = 1, \ldots, B, \tag{20}$$

where $\mathbf{E}_b \in \mathbb{R}^{L \times P}$, $\mathbf{A}_{f,b} \in \mathbb{R}^{P \times N_p}$, and $\hat{\mathbf{Y}}_{f,b} \in \mathbb{R}^{L \times N_p}$. Stacking the batchwise products gives $\hat{\mathbf{Y}}_f \in \mathbb{R}^{B \times L \times N_p}$.

E. Loss Function

GFMT-Net is trained end-to-end using a composite objective that promotes reconstruction fidelity, penalizes violations of the abundance sum-to-one constraint, and discourages endmember collapse:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{rec}} + \beta \mathcal{L}_{\text{ASC}} + \gamma \mathcal{L}_{\text{div}}, \tag{21}$$

where α, β, and γ control the relative contributions of the three terms.

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1) Reconstruction Loss: Let $\mathbf{Y}_f \in \mathbb{R}^{B\times L\times N_p}$ denote the input HSI flattened along its spatial dimensions, and let $\mathbf{y}_{b,n}$ and $\hat{\mathbf{y}}_{b,n}$ denote the observed and reconstructed spectra of pixel n in batch element b. The reconstruction loss combines the mean pixelwise SAD with a normalized squared reconstruction error:

$$\begin{aligned}\mathcal{L}_{\text{rec}} = &\frac{1}{BN_p}\sum_{b=1}^{B}\sum_{n=1}^{N_p}\operatorname{sad}(\mathbf{y}_{b,n},\hat{\mathbf{y}}_{b,n})\\ &+\frac{\lambda_{\text{mse}}}{BLN_p}\left\|\mathbf{Y}_f-\hat{\mathbf{Y}}_f\right\|_F^2,\end{aligned} \tag{22}$$

where

$$\operatorname{sad}(\mathbf{a},\mathbf{b}) = \arccos\left(\frac{\langle\mathbf{a},\mathbf{b}\rangle}{\|\mathbf{a}\|_2\|\mathbf{b}\|_2}\right), \tag{23}$$

and λ_{mse} balances spectral-shape and magnitude fidelity. The first term is invariant to a common positive rescaling of a spectrum, whereas the second penalizes absolute reconstruction errors.

2) Abundance Constraints: The final ReLU in the abundance head enforces the abundance nonnegativity constraint, $\mathbf{A} \geq 0$, directly. The abundance sum-to-one constraint is imposed by

$$\mathcal{L}_{\text{ASC}} = \frac{1}{BN_p}\sum_{b=1}^{B}\left\|\mathbf{1}_P^\top\mathbf{A}_{f,b}-\mathbf{1}_{N_p}^\top\right\|_2^2, \tag{24}$$

where $\mathbf{A}_{f,b}\in\mathbb{R}^{P\times N_p}$ is the flattened abundance matrix of batch element b. This term penalizes deviations of each pixel's abundance sum from one.

3) Diversity Loss: When multiple endmembers are estimated simultaneously, their spectra may collapse toward similar solutions. Let $\mathbf{e}_{b,i}\in\mathbb{R}^L$ denote the i-th endmember spectrum in batch element b. We apply the diversity regularization

$$\mathcal{L}_{\text{div}} = \frac{1}{BP(P-1)}\sum_{b=1}^{B}\sum_{i\neq j}\operatorname{ReLU}(\cos(\mathbf{e}_{b,i},\mathbf{e}_{b,j})-\tau), \tag{25}$$

where $\cos(\mathbf{a},\mathbf{b}) = \mathbf{a}^\top\mathbf{b}/(\|\mathbf{a}\|_2\|\mathbf{b}\|_2)$ and $\tau = 0.5$. Endmember pairs with cosine similarity below τ incur no penalty, whereas pairs above the threshold are penalized in proportion to the excess similarity.

IV. Experiments

We evaluate GFMT-Net on three real hyperspectral datasets. We first describe the experimental settings, including datasets, implementation details, comparison methods, and evaluation metrics. We then present quantitative comparisons with state-of-the-art methods, followed by component-wise ablation studies.

A. Experimental Settings

1) Datasets: We evaluate the proposed method on three real hyperspectral datasets.

Samson Dataset: The Samson dataset was acquired by the SAMSON sensor, containing 95×95 pixels with 156 spectral bands covering wavelengths from 401 to 889 nm. The spatial resolution is approximately 3.2 m. Three endmembers are identified: *Soil*, *Tree*, and *Water*.

Jasper Ridge Dataset: The Jasper Ridge dataset is a subimage of AVIRIS hyperspectral data with 100×100 pixels. After removing channels 1–3, 108–112, 154–166, and 220–224 owing to atmospheric absorption and low signal-to-noise ratios, 198 spectral bands covering 380–2500 nm are retained. Four endmembers are identified: *Tree*, *Water*, *Soil*, and *Road*.

Urban4 Dataset: The Urban4 dataset is extracted from the HYDICE Urban scene, containing 307×307 pixels with 162 spectral bands after removing channels with low signal-to-noise ratio. Four endmembers are identified: *Asphalt*, *Grass*, *Tree*, and *Roof*.

2) Implementation Details: The proposed GFMT-Net is implemented in PyTorch and trained on an NVIDIA GeForce RTX 4070 SUPER GPU. The GAC module learns location-dependent vertical and horizontal sampling scales through lightweight subnetworks, with the shared grid dimensions (k_h, k_w) determined by spatial averaging and odd-integer quantization. The grid dimensions are learned during the first 100 epochs and then frozen for the remaining training. We adopt the Adam optimizer with a multi-step learning rate scheduler across all three datasets. The initial learning rate, weight decay, decay interval, and total training epochs are set per dataset: 6×10^{-4}, 0.1, 500 epochs and 2000 epochs for Samson; 2×10^{-4}, 0.1, 245 epochs and 1300 epochs for Jasper Ridge; and 2×10^{-5}, 5×10^{-2}, 300 epochs and 1500 epochs for Urban4, respectively. The batch size is set to 1 for all datasets. For statistical stability and reproducibility, all results are averaged over ten independent runs with different random seeds.

For GLMT, we use two complete-sequence MambaVision mixers followed by two windowed Transformer blocks, with $W_s = 8$, 4 attention heads, and a latent dimension $C = 64$ (16 dimensions per head). The revised FSE branch uses adaptive pooling with $G = 10$, two FSE mappings with input, intermediate, and output dimensions of 100, 100, and 40, respectively, a periodic ratio $p_r = 0.25$, and a wavelength-axis convolution with kernel size 5. The loss weights (α,β,γ) are set to $(1.0, 0.015, 0.01)$ for Samson, $(0.5, 0.2, 0.2)$ for Jasper Ridge, and $(0.8, 0.01, 0.2)$ for Urban4. All experiments are conducted under identical hardware and software environments to ensure fair comparison.

For comparison, we select eight state-of-the-art hyperspectral unmixing methods, including CyCUNet [42], DAEN [18], A2SN [21], DFFN [43], MsCM-Net [22], TSUCAE-Net [44], SWC-Net [37], and DTU-Net [45]. All methods are evaluated under consistent experimental conditions using their publicly available implementations with the same training-testing splits and evaluation protocol.

3) Evaluation Metrics: We adopt two widely used metrics to quantify unmixing accuracy:

SAD: SAD measures the spectral similarity between the estimated endmember $\hat{\mathbf{e}}_p$ and the ground-truth endmember $\mathbf{e}_p$, defined as

$$\text{SAD}_p = \arccos\left(\frac{\hat{\mathbf{e}}_p^\top\mathbf{e}_p}{\|\hat{\mathbf{e}}_p\|_2\|\mathbf{e}_p\|_2}\right). \tag{26}$$

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TABLE I
SAD AND RMSE COMPARISON ON THE SAMSON DATASET

Metric	DAEN	CyCUNet	A2SN	DFFN	MsCM-Net	TSUCAE-Net	SWC-Net	DTU-Net	GFMT-Net
Soil	0.1566	0.0422	0.0597	0.0419	0.0245	0.0129	0.0472	0.0131	**0.0092**
Tree	0.0587	0.0399	0.0501	0.0374	0.0526	**0.0308**	0.0330	0.0313	0.0372
Water	0.1997	0.0326	0.0786	0.0208	0.0358	0.0290	**0.0090**	0.0271	0.0379
Mean SAD	0.1383	0.0382	0.0628	0.0334	0.0376	0.0243	0.0297	**0.0239**	0.0281
Soil	0.1856	0.1662	0.0609	**0.0256**	0.0712	0.0673	0.0545	0.0632	0.0377
Tree	0.1646	0.1692	0.0405	**0.0306**	0.0672	0.0612	0.0488	0.0428	0.0368
Water	0.0695	0.1332	0.0582	0.0516	**0.0307**	0.0321	0.0501	0.0480	0.0317
Mean RMSE	0.1399	0.1562	0.0532	0.0359	0.0563	0.0535	0.0512	0.0513	**0.0354**

A lower SAD value indicates better endmember estimation accuracy.

Root Mean Square Error (RMSE): RMSE evaluates the per-pixel abundance estimation error for the p-th endmember:

$$\mathrm{RMSE}_p = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{a}_{i,p} - a_{i,p})^2}, \tag{27}$$

where N is the total number of pixels, and $\hat{a}_{i,p}$ and $a_{i,p}$ denote the estimated and ground-truth abundances, respectively.

For both metrics, lower values indicate better performance. We report per-material SAD and RMSE along with the mean value across all endmembers. The best results are highlighted in bold and the second-best results are underlined in all comparison tables.

B. Experiments on the Samson Dataset

The Samson scene contains three materials distributed over both broad homogeneous regions and gradual transition areas. As shown in Table I, GFMT-Net obtains the lowest Mean RMSE of 0.0354, reducing the error by 1.4% relative to the second-best DFFN result of 0.0359. Although this margin is modest, GFMT-Net ranks second for the RMSE of every material, showing that its aggregate abundance accuracy does not arise from a single dominant result. For endmember estimation, GFMT-Net achieves the lowest Soil SAD of 0.0092, which is 28.7% lower than the second-best value of 0.0129 obtained by TSUCAE-Net. Its Mean SAD of 0.0281 ranks third, mainly because the Water SAD of 0.0379 is less competitive than its Soil and Tree results.

The endmember curves in Fig. 3 agree with the quantitative results: the Soil spectrum estimated by GFMT-Net closely follows the reference curve, whereas a larger discrepancy is visible for Water. In Fig. 4, GFMT-Net preserves the principal Tree and Water regions with continuous responses and recognizable boundaries, while several competing estimates exhibit either stronger local fluctuations or smoother transitions. These observations support the stable material-wise abundance performance in Table I, while also showing that endmember fidelity remains material dependent.

The stable abundance estimates benefit from the coordinated processing in the geometry pathway. GAC adjusts the vertical and horizontal sampling extents across homogeneous and transition regions, reducing the geometric mismatch imposed by a single fixed sampling grid. GLMT subsequently propagates information over the complete spatial sequence before refining local relations within windows, which promotes continuity in broad material regions while retaining their boundaries. In the frequency pathway, FSE enriches the bandwise spatial descriptors with sinusoidal and nonperiodic basis responses and then couples adjacent wavelengths, contributing to the close Soil spectral fit. The weaker Water SAD nevertheless indicates that the benefit of this spectral representation remains material dependent.

C. Experiments on the Jasper Ridge Dataset

Jasper Ridge contains irregular material regions together with narrow Road structures, making both regional coherence and fine-structure recovery important. Table II shows that GFMT-Net obtains the best result on five of the ten reported metrics. In particular, its Mean SAD of 0.0521 and Mean RMSE of 0.0727 are 12.9% and 21.4% lower, respectively, than the corresponding second-best results. The improvement in endmember estimation is especially pronounced for Tree, for which GFMT-Net achieves an SAD of 0.0350, 60.3% below the second-best TSUCAE-Net result of 0.0881; it also obtains the lowest Soil SAD. In contrast, its Water and Road SAD values do not rank among the top two, indicating that the spectral advantage is not uniform across materials.

The spectral curves in Fig. 5 provide a consistent visual comparison. GFMT-Net follows the characteristic red-edge transition of Tree closely, whereas the larger deviation of its Road spectrum agrees with the corresponding SAD. The abundance maps in Fig. 6 further show that GFMT-Net suppresses much of the background response in the Tree map and recovers the principal narrow Road segments, whereas several competing methods introduce scattered activation or broaden these structures. The agreement between the spectra, abundance maps, and aggregate metrics demonstrates that GFMT-Net maintains a favorable balance between regional abundance coherence and fine spatial recovery on this dataset.

This performance benefits from the complementary spatial roles of GAC and GLMT. The location-dependent axis-aligned sampling extents of GAC accommodate the coexistence of broad irregular regions and narrow spatial structures, while the complete-sequence MambaVision stage supplies cross-region context before windowed attention refines compact

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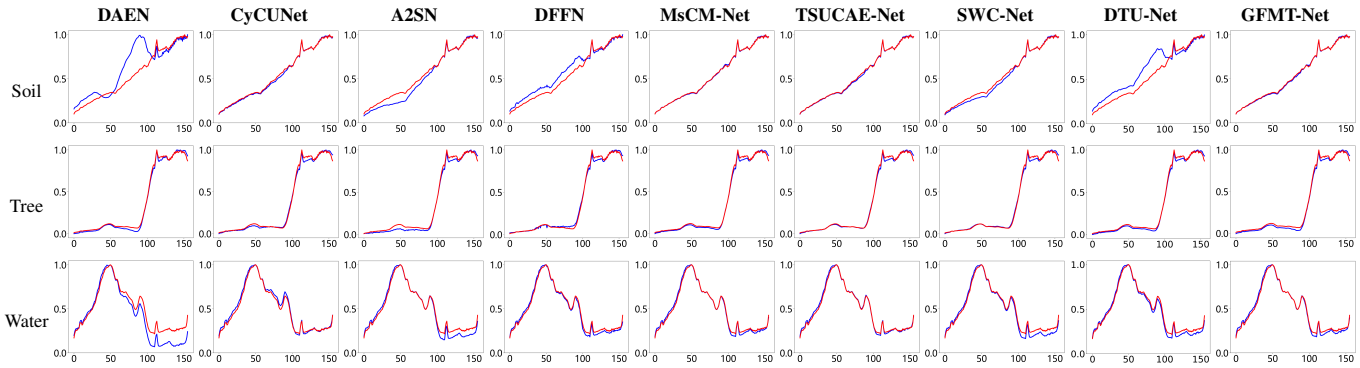

Fig. 3. Endmember spectra comparison on the **Samson** dataset. From left to right: DAEN, CyCUNet, A2SN, DFFN, MsCM-Net, TSUCAE-Net, SWC-Net, DTU-Net, and GFMT-Net. From top to bottom: Soil, Tree, Water. The red curves denote the reference ground-truth endmember spectra, and the blue curves are the estimated spectra.

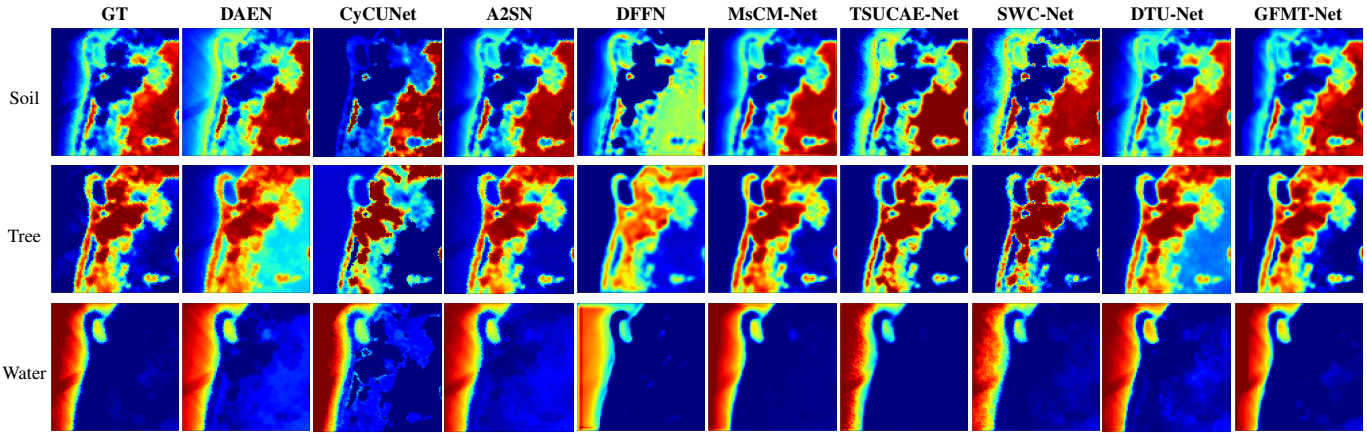

Fig. 4. Abundance map comparison on the **Samson** dataset. From left to right: Ground Truth (GT), DAEN, CyCUNet, A2SN, DFFN, MsCM-Net, TSUCAE-Net, SWC-Net, DTU-Net, and GFMT-Net. From top to bottom: Soil, Tree, Water.

TABLE II
SAD AND RMSE COMPARISON ON THE JASPER RIDGE DATASET

Metric	DAEN	CyCUNet	A2SN	DFFN	MsCM-Net	TSUCAE-Net	SWC-Net	DTU-Net	GFMT-Net
Tree	0.1791	0.1198	0.1167	0.1035	0.1609	0.0881	0.1280	0.1148	**0.0350**
Water	0.2537	0.0824	0.0866	0.0434	**0.0298**	0.0313	0.0703	0.0314	0.0600
Soil	0.2189	0.0935	0.2075	0.0645	0.0967	0.1554	0.1192	0.0666	**0.0468**
Road	0.4792	**0.0264**	0.1919	0.0580	0.0332	0.0861	0.0958	**0.0264**	0.0666
Mean SAD	0.2827	0.0805	0.1507	0.0674	0.0801	0.0902	0.1033	0.0598	**0.0521**
Tree	0.1127	0.0864	0.1880	0.0815	0.0975	0.1894	0.1011	0.1147	**0.0523**
Water	0.1267	0.1175	0.0917	0.0635	0.1500	**0.0295**	0.0457	0.0447	0.0297
Soil	0.1548	**0.1054**	0.2060	0.1162	0.1321	0.2337	0.1470	0.1272	0.1258
Road	0.1488	**0.0800**	0.1821	0.1088	0.1738	0.2043	0.1203	0.0988	0.0830
Mean RMSE	0.1358	0.0973	0.1670	0.0925	0.1383	0.1642	0.1035	0.0964	**0.0727**

local patterns. Their coordination helps reduce background interference without discarding the principal Road structures. For endmember recovery, the basis-enriched band descriptors and wavelength-axis convolution in FSE strengthen the representation of local spectral transitions, enabling GFMT-Net to follow the Tree red-edge more accurately. The contrast between the recovered Road structures and its less competitive Road SAD also shows that accurate spatial abundance recovery does not necessarily imply equally accurate endmember estimation for every material.

D. Experiments on the Urban4 Dataset

Urban4 presents a larger and more spatially complex scene in which natural materials and man-made surfaces form numerous adjacent regions and narrow boundaries. As reported in Table III, GFMT-Net achieves the second-lowest Mean RMSE of 0.1037 and obtains the lowest Grass and Roof RMSE values. Its endmember performance is less uniform: GFMT-Net gives the lowest Asphalt SAD of 0.0626, but its Mean SAD of 0.0610 ranks third behind MsCM-Net and TSUCAE-Net. The difference between the two aggregate rankings shows

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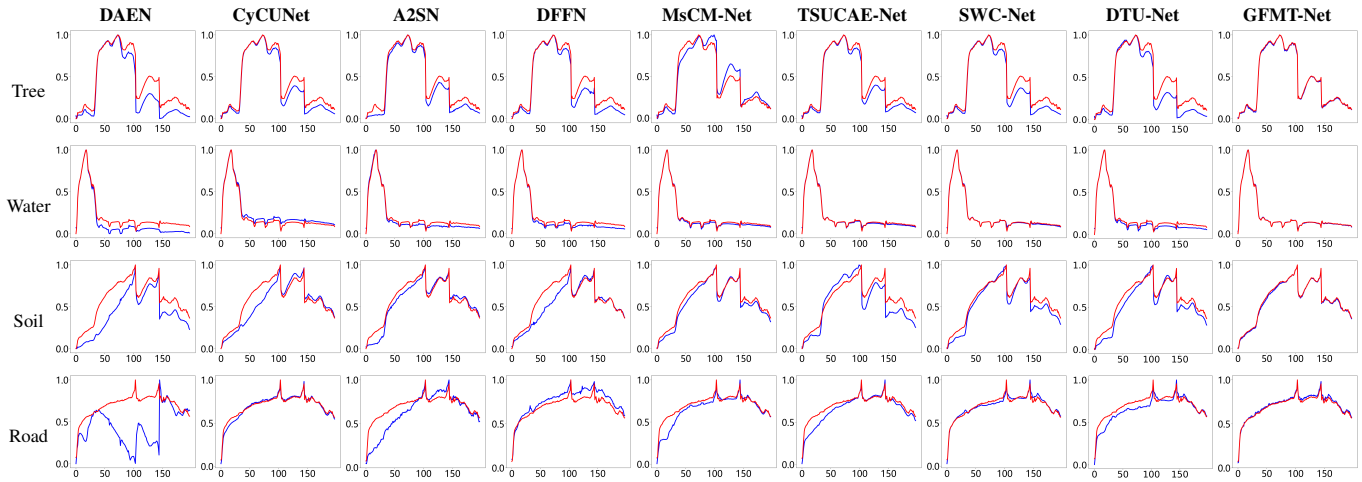

Fig. 5. Endmember spectra comparison on the **Jasper Ridge** dataset. From left to right: DAEN, CyCUNet, A2SN, DFFN, MsCM-Net, TSUCAE-Net, SWC-Net, DTU-Net, and GFMT-Net. From top to bottom: Tree, Water, Soil, Road. The red curves denote the reference ground-truth endmember spectra, and the blue curves are the estimated spectra.

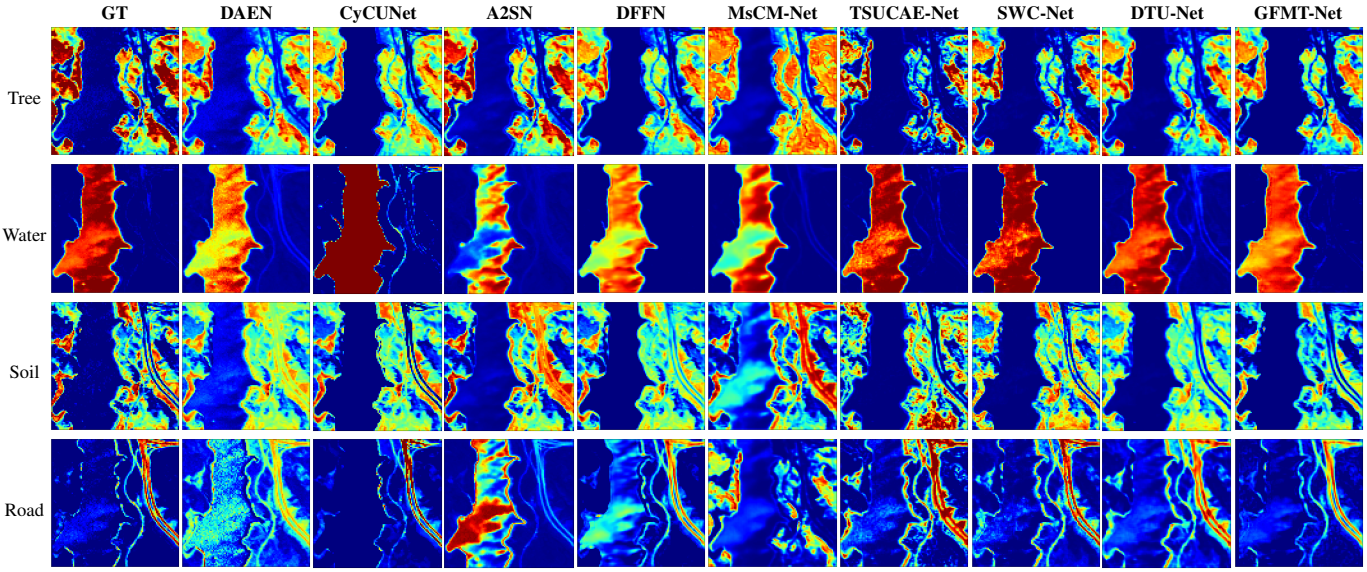

Fig. 6. Abundance map comparison on the **Jasper Ridge** dataset. From left to right: Ground Truth (GT), DAEN, CyCUNet, A2SN, DFFN, MsCM-Net, TSUCAE-Net, SWC-Net, DTU-Net, and GFMT-Net. From top to bottom: Tree, Water, Soil, Road.

that GFMT-Net remains competitive in abundance estimation on this scene, whereas its endmember accuracy varies more substantially across materials.

In Fig. 7, the Asphalt spectrum estimated by GFMT-Net follows the reference curve more closely than its estimates for several other materials, consistent with the material-wise SAD values. The abundance maps in Fig. 8 retain continuous responses over the principal Asphalt and Grass regions and localize the dominant Roof region, although ambiguity remains around some Tree and Roof boundaries. These qualitative patterns are consistent with the stronger RMSE ranking and the more variable SAD results reported in Table III.

The competitive abundance performance on this larger scene benefits from the geometry pathway at two complementary spatial ranges. GAC adapts the sampling extents to heterogeneous local layouts, whereas GLMT combines complete-sequence information propagation with window-level relation modeling to maintain scene-level consistency and local spatial detail. This coordination contributes to the continuous Asphalt and Grass responses and the localization of the principal Roof region. By contrast, the mixed SAD ranking shows that FSE does not yield uniform endmember gains across the spectrally diverse urban materials, despite the accurate Asphalt estimate.

Across the three real datasets, GFMT-Net achieves the lowest Mean RMSE on Samson and Jasper Ridge and the second-lowest on Urban4, demonstrating comparatively stable abundance estimation under different spatial organizations. Its Mean SAD ranks first on Jasper Ridge and third on Samson and Urban4, indicating greater dataset- and material-level variation in endmember recovery. This cross-dataset pattern reflects the division of labor in the dual-domain framework: GAC and GLMT provide adaptive local support and broad-to-local spatial interaction for abundance estimation, whereas FSE uses reconstruction-conditioned basis enrichment

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TABLE III
SAD AND RMSE COMPARISON ON THE URBAN4 DATASET

Metric	DAEN	CyCUNet	A2SN	DFFN	MsCM-Net	TSUCAE-Net	SWC-Net	DTU-Net	GFMT-Net
Asphalt	0.2028	0.1385	0.1209	0.1926	0.0659	0.0682	0.0869	0.0695	**0.0626**
Grass	0.1554	0.3455	0.1038	0.0752	0.0444	**0.0347**	0.0468	0.0832	0.0543
Tree	0.1184	0.3877	0.0409	0.0828	**0.0298**	0.0472	0.1067	0.0709	0.0616
Roof	0.4506	0.0618	0.0904	0.0810	**0.0260**	0.0489	0.1302	0.0448	0.0655
Mean SAD	0.2318	0.2334	0.0890	0.1079	**0.0415**	0.0498	0.0926	0.0671	0.0610
Asphalt	0.1302	0.2845	**0.1124**	0.1196	0.1430	0.1395	0.1130	0.1721	0.1315
Grass	0.1768	0.2901	0.1173	0.1461	0.1386	0.1178	0.1130	0.1390	**0.1033**
Tree	0.1320	0.2893	**0.0657**	0.1312	0.1049	0.0754	0.1021	0.0834	0.0870
Roof	0.1163	0.1363	0.0992	0.1504	0.0948	0.1068	0.0938	0.1089	**0.0932**
Mean RMSE	0.1388	0.2500	**0.0986**	0.1368	0.1203	0.1099	0.1055	0.1259	0.1037

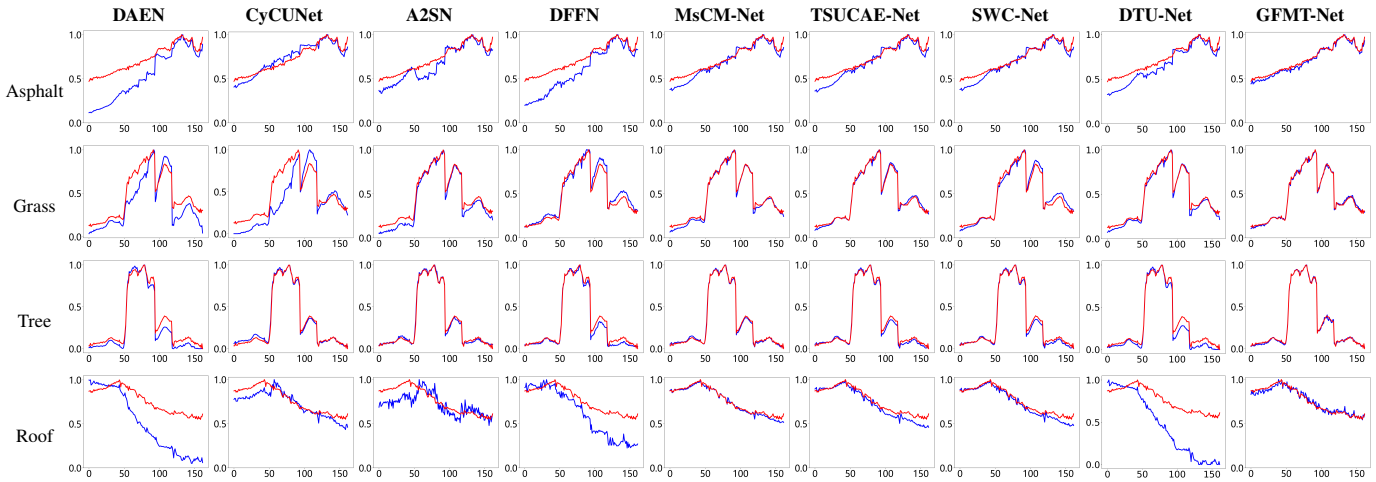

Fig. 7. Endmember spectra comparison on the **Urban4** dataset. From left to right: DAEN, CyCUNet, A2SN, DFFN, MsCM-Net, TSUCAE-Net, SWC-Net, DTU-Net, and GFMT-Net. From top to bottom: Asphalt, Grass, Tree, Roof. The red curves denote the reference ground-truth endmember spectra, and the blue curves are the estimated spectra.

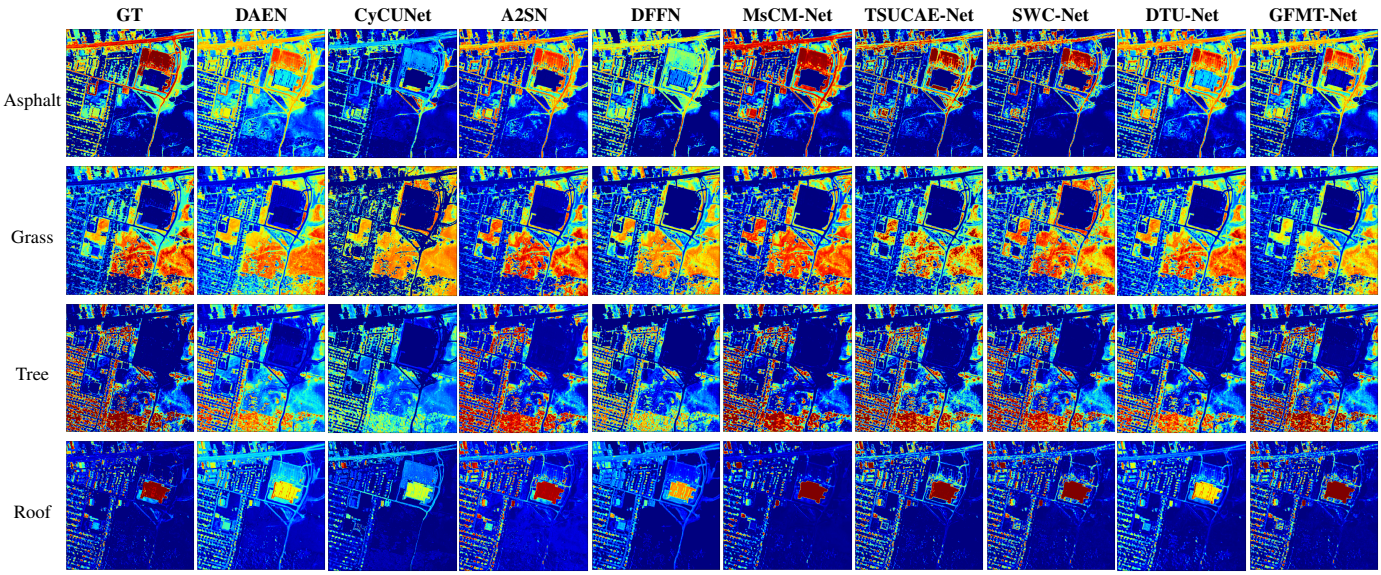

Fig. 8. Abundance map comparison on the **Urban4** dataset. From left to right: Ground Truth (GT), DAEN, CyCUNet, A2SN, DFFN, MsCM-Net, TSUCAE-Net, SWC-Net, DTU-Net, and GFMT-Net. From top to bottom: Asphalt, Grass, Tree, Roof.

and local wavelength coupling for endmember recovery. The controlled replacements in Table VII further support this interpretation. Replacing GAC with a standard convolution, reorganizing GLMT as either a Mamba-only or Transformer-

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TABLE IV
PARAMETER SENSITIVITY ANALYSIS ON THE SAMSON DATASET

Window Size W_s				
Metric	**4**	**8**	**12**	**16**
Mean SAD	0.0298	**0.0281**	0.0359	0.0319
Mean RMSE	**0.0352**	0.0354	0.0551	0.0361
Periodic Ratio p_r				
Metric	**0.15**	**0.25**	**0.35**	**0.45**
Mean SAD	0.0425	**0.0281**	0.3100	0.3745
Mean RMSE	0.0455	**0.0354**	0.1716	0.1962

only backbone, or replacing the basis-enriched FSE mappings with GELU-activated linear layers while retaining adaptive pooling and wavelength-axis convolution increases both Mean SAD and Mean RMSE on Samson and Jasper Ridge. Together, the comparison and ablation results support the contributions of geometry-adaptive sampling, the mixed Mamba–Transformer organization of GLMT, and the basis-enriched mappings within FSE, while showing that the abundance gains are more consistent across datasets than the endmember gains.

E. Parameter Sensitivity Analysis

We analyze sensitivity to two key hyperparameters: the window size W_s in the GLMT backbone and the periodic ratio p_r in the FSE layers, on the Samson dataset. Samson provides a moderate scene size (95×95 pixels), 156 spectral bands, and three endmembers, offering representative spatial–spectral complexity while keeping the repeated hyperparameter scans computationally tractable. We therefore use it as a controlled setting for isolating the effects of W_s and p_r. This single-dataset analysis is not intended to establish universal optima; the selected architectural settings are subsequently kept fixed in the Jasper Ridge and Urban4 evaluations, while their dataset-specific training schedules and loss weights remain unchanged. Results are reported in Table IV.

Window Size W_s in GLMT: The window size controls the spatial extent of self-attention. We evaluate $W_s \in \{4, 8, 12, 16\}$. The lowest Mean SAD is obtained at $W_s = 8$ (0.0281), whereas the lowest Mean RMSE is obtained at $W_s = 4$ (0.0352), followed closely by $W_s = 8$ (0.0354). The results are non-monotonic across the tested settings and therefore do not support a claim that one window size is uniformly optimal for both metrics.

Periodic Ratio p_r in FSE: The periodic ratio determines the proportion of FSE outputs allocated to cosine and sine components. Among $p_r \in \{0.15, 0.25, 0.35, 0.45\}$, $p_r = 0.25$ gives the lowest values for both metrics in the current results. The large errors at 0.35 and 0.45 show sensitivity to this ratio, but the experiment alone does not identify the underlying optimization or representation mechanism.

F. Noise Robustness Analysis

To examine the behavior of GFMT-Net under spectral corruption, we conduct an additional robustness experiment on Samson. Zero-mean Gaussian noise is added to the normalized HSI at nominal SNR levels of 30, 20, and 10 dB, after which the noisy observations are clipped to $[0, 1]$. The model is evaluated over ten runs at each noise level using the same optimization protocol as for the clean data, and Table V reports the mean results.

TABLE V
NOISE ROBUSTNESS ANALYSIS ON THE SAMSON DATASET

Metric	**Clean**	**30 dB**	**20 dB**	**10 dB**
Mean SAD	**0.0281**	0.0367	0.0294	0.0384
Mean RMSE	**0.0354**	0.0618	0.0468	0.0864

TABLE VI
EFFECT OF GLOBAL AND LOCAL DEPTHS IN GLMT ON SAMSON

Global	**Local**	**Mean SAD**	**Mean RMSE**	**Params (M)**
4	4	0.0412	0.0416	2.578
4	0	0.0311	0.0387	2.332
0	4	0.0368	0.0664	2.406
2	2	**0.0281**	**0.0354**	2.393
2	0	0.0477	0.0546	2.293
0	2	0.0453	0.0448	2.307

Both errors are higher under every noisy condition than on the clean Samson data, with the largest degradation observed at 10 dB. The results are not strictly monotonic with respect to the nominal SNR because the 20-dB condition performs better than the 30-dB condition. We therefore interpret this experiment as characterizing the empirical behavior of GFMT-Net at representative noise levels, rather than as evidence of monotonic degradation. Even under the strongest tested corruption, the model remains trainable and produces finite unmixing estimates, although the increased Mean RMSE indicates that abundance estimation is more sensitive to severe noise than in the clean setting.

G. GLMT Depth Analysis

We further study the effect of the numbers of global MambaVision mixers and local windowed Transformer blocks on Samson. In Table VI, $n_g + n_l$ denotes a configuration with n_g global mixers followed by n_l local blocks, and zero indicates that the corresponding stage is omitted. All configurations use the same data and optimization settings, and their results are averaged over ten runs. The $4 + 0$ and $0 + 4$ entries use the same configurations and results as the w/o Transformer and w/o Mamba variants, respectively, in Table VII.

Among the evaluated configurations, the default $2 + 2$ organization achieves the lowest Mean SAD and Mean RMSE. Increasing both stages to $4+4$ raises the parameter count from 2.393 M to 2.578 M without improving either metric. The global-only and local-only configurations also do not match the joint $2 + 2$ result. These observations support two global mixers followed by two local blocks as the default empirical choice among the tested depths, while avoiding a claim of global optimality beyond the evaluated configurations.

H. Ablation Study

To evaluate the contribution of each component, we compare four controlled variants with the full model on Samson and Jasper Ridge:

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TABLE VII
ABLATION STUDY ON SAMSON AND JASPER RIDGE DATASETS

Variant	Samson		Jasper Ridge	
	Mean SAD	Mean RMSE	Mean SAD	Mean RMSE
w/o GAC	0.0344	0.0476	0.0698	0.1015
w/o Transformer	0.0311	0.0387	0.0580	0.0861
w/o Mamba	0.0368	0.0664	0.0615	0.1049
w/o FSE	0.0413	0.0622	0.0680	0.0998
GFMT-Net (Full)	**0.0281**	**0.0354**	**0.0521**	**0.0727**

1) **w/o GAC:** Replacing the GAC module with a standard 3×3 convolutional layer. This variant uses fixed square receptive fields for all spatial locations.
2) **w/o Transformer:** Replacing the windowed self-attention blocks with additional MambaVision blocks, thereby forming a Mamba-only GLMT backbone with matched depth and a similar parameter count.
3) **w/o Mamba:** Replacing the MambaVision blocks with additional windowed self-attention blocks, thereby forming a Transformer-only GLMT backbone with matched depth and a similar parameter count.
4) **w/o FSE:** Replacing only the two sinusoidal/nonperiodic basis mappings with linear layers followed by GELU. Adaptive pooling and the wavelength-axis convolution are retained in this variant.
5) **GFMT-Net (Full):** The complete proposed model integrating all three components with the two-stage SSM-Transformer backbone.

Table VII summarizes the results. Replacing GAC with a standard convolution increases both Mean SAD and Mean RMSE on Samson and Jasper Ridge, demonstrating the value of location-dependent spatial sampling. The w/o Transformer and w/o Mamba settings should be interpreted as alternative single-mixer-type organizations of GLMT rather than literal module-deletion tests. The Mamba-only organization yields lower errors than the Transformer-only organization on both datasets, while the full mixed backbone performs better than either alternative. These comparisons support the use of complete-sequence MambaVision modeling together with subsequent windowed local refinement under the evaluated settings. On Samson, the w/o Transformer, w/o Mamba, and full configurations contain 2.332, 2.406, and 2.393 million parameters, respectively; the corresponding counts on Jasper Ridge are 5.891, 6.023, and 6.008 million. Their comparable sizes indicate that the observed performance differences are unlikely to be explained by parameter count alone. Replacing the FSE basis mappings with GELU-activated linear layers also increases both metrics. Because adaptive pooling and the wavelength-axis convolution remain unchanged, this comparison specifically supports the value of the sinusoidal and nonperiodic basis responses rather than the entire endmember branch. These controlled replacements support the complementary roles of location-dependent sampling in GAC, mixed global–local modeling in GLMT, and basis-enriched descriptor mapping in FSE.

V. CONCLUSION

This paper proposed GFMT-Net, a geometry–frequency dual-domain network for hyperspectral unmixing. In the geometry domain, GAC learns location-dependent sampling scales to adapt spatial feature extraction to heterogeneous material distributions, while GLMT combines complete-sequence Mamba modeling with windowed self-attention to promote region-level abundance coherence and preserve fine spatial structures. Based on the resulting internal reconstruction, FSE enriches per-band spatial descriptors with sinusoidal and nonperiodic basis responses and exploits local dependencies along the ordered wavelength sequence, thereby improving endmember representation in the frequency domain. Experimental results on three real hyperspectral datasets demonstrate the competitive performance of GFMT-Net in both abundance and endmember estimation, while the controlled variants support the contributions of geometry-adaptive sampling, the mixed Mamba–Transformer organization, and the basis-enriched FSE mappings. Future work will investigate its robustness across more diverse scenes and spectral variability conditions.

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